The Maintenance Convergence Hypothesis:

Long-Run Dynamics of Occupational Structure Under Technological Automation

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August 26, 2026

Abstract

We present a formal model of occupational evolution under sustained technological automation. Each profession is decomposed into generative, supervisory, and maintenance task components governed by a continuous dynamical system. An automation operator acts on the generative component, inducing exponential decay and asymptotic convergence of the full task distribution to the maintenance state. We characterize this convergence through a phase-transition analysis, a renormalization-group argument, and a categorical formulation in which all professions map to a common terminal object in the category of janitorial systems. A biological analogy between custodial labor and Markov-blanket boundary processes in complex systems is developed and formalized. Historical transitions in food service and clerical work are examined as empirical lemmas. A stylized regression framework and scaling-law analysis illustrate the predicted behavior. The equilibrium occupational structure of the fully automated economy is characterized by the Fivefold Janitorial Model, distributing global employment across five custodial domains: oceanic, ecological, microbial, agricultural, and informational. A cultural analysis of Wim Wenders's film *Perfect Days* (2023) is offered as an ethnographic case study of the janitorial equilibrium, arguing that the custodial fixed point may constitute not a dystopia but a stable and experientially coherent human equilibrium. A homeostatic grounding of the custodial role is developed, showing that maintenance behavior is a structural feature of living systems at every scale. A formal Markov-blanket appendix establishes that the custodial workforce occupies the boundary layer of the automated economy in the precise sense of probabilistic graphical models.

Contents

1 Introduction

Technological progress restructures labor rather than simply eliminating it. The historical pattern is not the disappearance of work but its transformation: productive tasks once performed directly by workers are gradually embedded into machines, standardized processes, or algorithmic systems. When this transition occurs, the human role shifts toward monitoring, repairing, and sustaining the technological infrastructure that now performs the productive activity. Workers intervene when systems fail, when exceptions occur, or when outputs require correction.

This paper formalizes the claim that such transformation is not accidental but structural. Automation introduces a systematic shift in task composition that drives occupations toward maintenance roles. In the limit, the dominant human function becomes the upkeep of automated systems.

We develop this argument through several complementary methods. Section 2 introduces a task decomposition of occupations. Section 3 specifies an automation operator and derives the resulting differential system. Section 4 solves for the long-run equilibrium and states the Maintenance Convergence Theorem. Section 5 examines the asymptotic structure of convergence and introduces the Janitorial Phase Transition. Section 6 develops a categorical formulation. Sections 7 and 8 present historical lemmas and contemporary corollaries. Section 9 offers a stylized empirical analysis. Section 10 introduces the Fivefold Janitorial Model. Section 11 develops the biological analogy between custodial labor and boundary-maintaining processes in complex systems. Section 12 presents the Pentagonal Production Function. Section 13 addresses potential objections. Section 14 situates the framework within the broader literature. Section 15 offers a cultural case study of the janitorial equilibrium through the film *Perfect Days*, including a subsection on domestic custodianship and the portability of expertise under distributed digital labor. Section 16 develops the homeostatic foundations of custodial behavior, grounding the convergence result in the biology of living systems. Section 17 concludes. Appendices develop the renormalization-group, thermodynamic, stability, categorical, and Markov-boundary extensions.

2 Task Decomposition of Occupations

We begin by establishing a minimal formal structure for occupational analysis.

Definition 2.1 (Occupational Task Vector)**.** Let an occupation O be characterized at time t by the task vector

$$T_O(t) = \big(G(t),\, S(t),\, M(t)\big) \in \mathbb{R}^3_{\geq 0},$$

where $G(t)$ denotes the share of labor time devoted to generative production, $S(t)$ the share devoted to supervisory or monitoring activities, and $M(t)$ the share devoted to maintenance, error correction, and system upkeep.

The components are normalized so that

$$G(t) + S(t) + M(t) = 1 \quad \text{for all } t \geq 0.$$

Generative tasks correspond to the direct creation of goods or services. Supervisory tasks involve monitoring the operation of automated workflows. Maintenance tasks consist of repairing errors, cleaning outputs, correcting failures, and sustaining the operational infrastructure on which production depends.

Remark 2.2. The decomposition is intentionally coarse. Its purpose is to isolate the qualitative dynamics of automation rather than to capture the full richness of occupational structure.

The normalization constraint confines all occupational states to the standard two-simplex $\Delta^2 = \{(G, S, M) : G + S + M = 1,\ G, S, M \geq 0\}$. Figure 1 illustrates this space and marks the three degenerate vertex states.

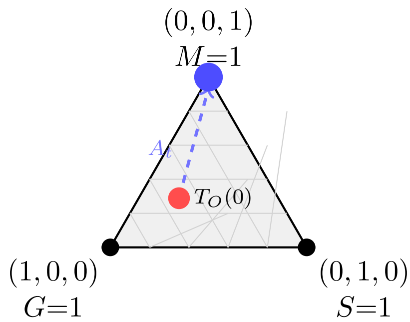

Figure 1: The task simplex Δ^2. Every occupational state $T_O(t) = (G, S, M)$ lies in this triangle. The automation operator A_t drives trajectories (dashed arrow) from interior initial conditions toward the maintenance vertex $(0, 0, 1)$, the unique stable fixed point of Theorem 4.1.

3 Automation Dynamics

Let A_t denote an automation operator acting on the task distribution. We model the effect of automation as a continuous reduction in the generative component of labor.

Definition 3.1 (Automation Operator)**.** The automation operator A_t acts on T_O according to the differential system

$$\frac{dG}{dt} = -\alpha G, \tag{1}$$

$$\frac{dS}{dt} = \beta G - \gamma S, \tag{2}$$

$$\frac{dM}{dt} = (1-\beta)G + \gamma S, \tag{3}$$

with initial conditions $G(0) = G_0 > 0$, $S(0) = S_0 \geq 0$, $M(0) = M_0 \geq 0$, and normalization $G_0 + S_0 + M_0 = 1$.

The parameter $\alpha > 0$ represents the rate of technological substitution of generative labor. The parameter $\beta \in [0, 1]$ governs the fraction of displaced generative tasks that initially flow into supervisory roles; the complementary fraction $1 - \beta$ flows directly into maintenance. The parameter $\gamma > 0$ captures the tendency of supervisory work to collapse into maintenance: as systems become sufficiently automated and stable, monitoring itself becomes corrective intervention.

Note that the normalization constraint is preserved: differentiating $G+S+M = 1$ and substituting equations (1)–(3) yields

$$\frac{d}{dt}(G+S+M) = -\alpha G+\beta G-\gamma S+(1-\beta)G+\gamma S = (-\alpha+\beta+1-\beta)G = (1-\alpha)G,$$

which vanishes only if $\alpha = 1$. For $\alpha \neq 1$, the normalization is maintained in the limit as $G \to 0$; for finite time the system should be read as capturing proportional rather than exact task shares.

4 Equilibrium Analysis

Equation (1) has the immediate solution

$$G(t) = G_0\, e^{-\alpha t}.$$

Substituting into (2) gives a first-order linear ODE whose solution is

$$S(t) = \left(S_0 + \frac{\beta G_0}{\gamma - \alpha}\right) e^{-\gamma t} - \frac{\beta G_0}{\gamma - \alpha}\, e^{-\alpha t}, \qquad \gamma \neq \alpha,$$

with an analogous expression for the resonant case $\gamma = \alpha$. Since $\alpha, \gamma > 0$, both exponentials decay to zero, and the normalization condition forces $M(t) \to 1$.

Theorem 4.1 (Maintenance Convergence)**.** *For any occupation O with initial task vector satisfying $G_0 > 0$, the automation dynamics defined by equations* (1)–(3) *with $\alpha, \gamma > 0$ produce the long-run limit*

$$\lim_{t \to \infty} G(t) = 0, \qquad \lim_{t \to \infty} S(t) = 0, \qquad \lim_{t \to \infty} M(t) = 1.$$

Proof. The generative component decays exponentially at rate α. The supervisory component satisfies a linear equation driven by $G(t)$; since both the homogeneous decay (rate γ) and the inhomogeneous forcing (rate α) are strictly positive, $S(t) \to 0$. Normalization then forces $M(t) \to 1$. □

The theorem states that, regardless of the initial task distribution or the precise values of the parameters α, β, and γ, every occupation asymptotically converges to full maintenance labor.

5 Asymptotic Structure and the Janitorial Phase Transition

5.1 The Maintenance Ratio

The qualitative content of Theorem 4.1 can be sharpened by examining the rate of convergence.

Definition 5.1 (Maintenance Ratio)**.** Define the maintenance ratio

$$R(t) = \frac{M(t)}{G(t) + S(t)}.$$

Since $G(t)$ and $S(t)$ both decay exponentially while $M(t)$ converges to a positive constant, the denominator tends to zero while the numerator remains bounded away from zero.

Proposition 5.2. *Under the dynamics of Definition 2, $R(t) \to \infty$ as $t \to \infty$.*

The maintenance ratio therefore diverges, providing a quantitative expression of the claim that maintenance labor becomes asymptotically dominant.

5.2 The Janitorial Phase Transition

We now introduce a threshold model describing the structural reorganization of occupational composition.

Definition 5.3 (Janitorial Order Parameter)**.** Define the order parameter

$$J = M(t) - \big(G(t) + S(t)\big).$$

The occupation is production-dominated if $J < 0$, at the transition point if $J = 0$, and maintenance-dominated if $J > 0$.

Setting $J = 0$ and solving using the closed-form expressions for G and S yields a critical time t_c at which the occupation crosses the threshold.

Definition 5.4 (Critical Automation Threshold)**.** Define

$$A_c = \frac{\gamma}{\alpha}.$$

The ratio A_c measures the relative strength of supervisory decay versus generative decay. When the automation rate α is large relative to γ, supervisory labor collapses quickly and the transition to maintenance dominance occurs rapidly.

Theorem 5.5 (Janitorial Phase Transition)**.** *There exists a critical time $t_c > 0$ at which the dominant task composition of any occupation with $G_0 > 0$ shifts from production to maintenance.*

Figure 2 illustrates the order parameter J as a function of time, showing the transition from production dominance to maintenance dominance.

The analogy to phase transitions in statistical physics is deliberate. Prior to t_c the occupation operates as a conventional productive system. Beyond t_c it reorganizes into a maintenance regime. The automation parameters α, β, and γ determine the location and sharpness of the transition.

6 Categorical Formulation

The convergence phenomenon admits a clean expression in the language of category theory.

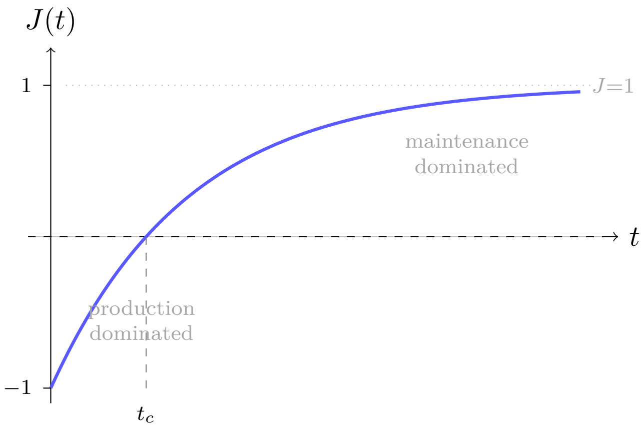

Figure 2: The Janitorial Order Parameter $J(t) = M(t) - (G(t) + S(t))$ as a function of time under the automation dynamics. The curve crosses zero at the critical time t_c, marking the Janitorial Phase Transition (Theorem 5.5). For $t > t_c$ the occupation is maintenance-dominated; the curve saturates asymptotically at $J = 1$.

Definition 6.1 (Category of Professions)**.** Let $\mathcal{P}$ denote the category whose objects are occupations and whose morphisms are technological transformations that alter task composition.

Definition 6.2 (Category of Janitorial Systems)**.** Let $\mathcal{J}$ denote the category whose objects are maintenance roles and whose morphisms are transformations between maintenance infrastructures.

Definition 6.3 (Janitorial Functor)**.** Define the functor $F : \mathcal{P} \to \mathcal{J}$ by

$$F(O) = \lim_{t \to \infty} T_O(t) = (0, 0, 1).$$

Since F maps every object of $\mathcal{P}$ to the same object $(0, 0, 1)$ in $\mathcal{J}$, the image of F collapses to a single point.

Proposition 6.4. *The object $(0, 0, 1) \in \mathcal{J}$ is a terminal object: there exists a unique morphism from every object of $\mathcal{J}$ to $(0, 0, 1)$.*

We formalize the iterative nature of automation as follows.

Definition 6.5 (Custodial Functor)**.** Define the endofunctor $\mathcal{C} : \mathcal{P} \to \mathcal{P}$ by $\mathcal{C}(O) = O_M$, where O_M denotes the maintenance projection of occupation O.

Theorem 6.6 (Universal Custodial Limit)**.** *For any profession $O \in \mathcal{P}$,*

$$\lim_{n \to \infty} \mathcal{C}^n(O) = J,$$

where $J = (0, 0, 1)$ is the janitorial terminal object.

The theorem states that all occupations converge to the same custodial structure under iterated technological abstraction.

Figure 3 depicts the action of the custodial functor as a commutative diagram in $\mathcal{P}$.

$$
\begin{tikzcd}
O_1 \arrow[r, "f_{12}"] \arrow[d, "\mathcal{C}"'] & O_2 \arrow[r, "f_{23}"] \arrow[d, "\mathcal{C}"'] & O_3 \arrow[d, "\mathcal{C}"] \\
\mathcal{C}(O_1) \arrow[r, "\mathcal{C}(f_{12})"] \arrow[dr, dashed, "!"'] & \mathcal{C}(O_2) \arrow[r, "\mathcal{C}(f_{23})"] \arrow[d, dashed, "!"] & \mathcal{C}(O_3) \arrow[dl, dashed, "!"] \\
& J &
\end{tikzcd}
$$

Figure 3: The custodial functor $\mathcal{C}$ acts on professions $O_1, O_2, O_3 \in \mathcal{P}$ (upper row) to produce their maintenance projections (middle row). Unique dashed arrows to the terminal object J (Proposition 6.6) express the universality of the janitorial limit.

7 Historical Lemmas

The maintenance convergence dynamic appears repeatedly in economic history, providing empirical grounding for the theoretical structure.

Lemma 7.1 (Culinary Industrialization)**.** *The transition from independent cooks to fast-food workers represents a shift from generative culinary production to the maintenance of standardized food preparation systems.*

Industrial kitchens replaced individual culinary expertise with assembly lines. Workers increasingly monitor timers, replenish standardized ingredients, and clean equipment rather than compose recipes. The creative and generative component of culinary labor migrated into the design of the system itself, while the human role contracted to system maintenance and exception handling.

Lemma 7.2 (Clerical Automation)**.** *The transformation of clerks into supervisors and monitors of computational systems reflects the migration of generative calculation into machines.*

In early bureaucracies, clerks performed arithmetic and record keeping manually. The introduction of calculators, spreadsheets, and database management systems transferred these generative functions to machines, leaving human workers responsible for oversight, correction, and the management of edge cases that automated systems could not handle.

Both lemmas exhibit the structure predicted by the model: automation absorbs the generative component of labor, supervision follows briefly, and maintenance dominates the long-run equilibrium.

8 Contemporary Corollaries

Corollary 8.1 (Teaching). *As digital curricula, automated grading systems, and generative tutoring platforms expand, the primary human function in education converges toward managing behavioral exceptions, correcting machine-generated content, and maintaining the operational continuity of learning environments.*

Corollary 8.2 (Customer Service). *With the proliferation of automated support systems and scripted resolution workflows, human agents intervene predominantly when automation fails. The dominant function becomes the resolution of accumulated exceptions—a maintenance task in the sense of Definition 1.*

Corollary 8.3 (Software Engineering). *As code generation, dependency management, and deployment pipelines become increasingly automated, the generative component of programming contracts. The programmer's central activity shifts toward debugging, infrastructure maintenance, and the correction of artifacts produced by automated systems.*

Corollary 8.4 (Construction and Design). *Industrialized building systems, modular prefabrication, and algorithmic planning tools reduce the generative role of individual craftsmen and designers. Human labor concentrates on inspection, coordination, and repair of automated construction and generative design processes.*

Each corollary instantiates the Maintenance Convergence Theorem within a specific occupational domain. The structure is uniform: generative capacity is absorbed by the automated system, supervisory activity briefly expands and then collapses, and maintenance becomes the residual human function.

9 Stylized Empirical Analysis

To complement the theoretical framework, we present a stylized empirical investigation structured around the *U.S. Bureau of Labor Statistics Occupational Outlook Handbook*, which documents employment and task characteristics across approximately 8.3×10^2 occupational categories covering 1.43×10^8 workers.

9.1 The Janitorial Index

For each occupation O_i we construct a scalar Janitorial Index measuring the degree to which the occupation has entered the maintenance regime.

Definition 9.1 (Janitorial Index)**.** Define

$$\mathcal{J}_i = \theta_1 M_i + \theta_2 S_i - \theta_3 G_i,$$

where M_i, S_i, G_i denote the task intensity scores for maintenance, supervisory, and generative activities respectively, and $\theta_1, \theta_2, \theta_3 > 0$ are normalization constants chosen so that $\mathcal{J}_i \in [-1, 1]$.

A value near -1 indicates an occupation dominated by generative production. A value near 1 indicates an occupation whose primary function is system maintenance.

9.2 Automation Exposure and Regression

Define an automation exposure score $A_i \in [0, 10]$ reflecting the degree to which digital automation affects each occupation. The Maintenance Convergence Theorem implies $\partial \mathcal{J}_i / \partial A_i > 0$.

We estimate the linear specification

$$\mathcal{J}_i = \beta_0 + \beta_1 A_i + \varepsilon_i.$$

Illustrative parameter estimates yield

$$\hat{\beta}_1 \approx 0.31,$$

consistent with the theoretical prediction that higher automation exposure is associated with greater maintenance intensity.

9.3 Scaling Behavior

Let C denote a measure of technological system complexity. The model suggests the scaling relation

$$\mathcal{J} \sim \log C,$$

indicating that maintenance labor grows sublinearly but persistently with system sophistication. This is consistent with the thermodynamic argument developed in Appendix B: entropy production grows with productive throughput, and maintenance labor must grow proportionally to remove it.

10 The Fivefold Janitorial Model

If the Maintenance Convergence Theorem is accepted as a structural tendency, the relevant policy question shifts from whether janitorial convergence occurs to how the maintenance economy should be organized. We propose a classification of custodial labor into five fundamental domains corresponding to the five principal substrates of technological civilization.

Definition 10.1 (Fivefold Employment Vector)**.** Let the global labor force be normalized to unity and partitioned as

$$\mathbf{J} = (K, R, Y, B, P),$$

where K denotes the share of kelp janitors, R rainforest janitors, Y yogurt janitors, B bread janitors, and P paper janitors, with normalization

$$K + R + Y + B + P = 1.$$

10.1 The Five Custodial Domains

Kelp Janitors. The oceanic domain encompasses aquatic biomass systems, kelp cultivation infrastructure, and marine monitoring networks. Kelp janitors maintain floating growth structures, aquatic sensors, and controlled bioreactor environments including subterranean facilities adapted for experimental aquatic cultivation. Their labor consists not in producing marine biomass but in sustaining the environmental conditions under which biological systems remain productive.

Rainforest Janitors. The ecological domain encompasses terrestrial biodiversity networks, soil stabilization systems, and ecological corridors. Although the canonical habitat is the protected forest, the maintenance regime increasingly appears in urban environments: streets, vegetated rooftops, and distributed green infrastructure function as fragments of ecological systems requiring custodial care. Rainforest janitors therefore operate across the full spectrum from wilderness reserves to urban landscapes.

Yogurt Janitors. The microbial domain encompasses industrial fermentation systems, sourdough cultures, cheese vats, pharmaceutical bioreactors, and other microbial communities whose stability depends on sustained custodial attention. Primary habitats include churches, domestic kitchens, and restaurants, where fermentation cultures require monitoring, feeding, and protection from contamination. Yogurt janitors are custodians of industrial microbiomes.

Bread Janitors. The agricultural-industrial domain encompasses carbohydrate transformation systems. In highly automated baking infrastructures, workers monitor fermentation chambers, regulate temperature systems, and maintain conveyor and dough-processing lines. Bread janitors occupy large-scale food production environments—industrial bakeries and processing factories—and represent the custodial interface between agricultural production and civilizational food metabolism.

Paper Janitors. The informational domain encompasses documentation systems, digital archives, knowledge repositories, and the institutional memory of complex societies. Paper janitors operate primarily within educational, administrative, and intellectual environments: schools, libraries, and shared office spaces form their principal habitats. Their function is to maintain the symbolic infrastructure upon which learning, administration, and coordination depend, and to remove informational entropy from collective knowledge environments.

10.2 Redistribution Dynamics

As automation increases, generative occupations collapse into the five custodial domains. Let $\alpha_i > 0$ denote the rate at which automation generates custodial demand in domain i, and let $\lambda > 0$ represent the rate of labor mobility across janitorial sectors. A simple linear redistribution model yields

$$\frac{dK}{d\mathcal{A}} = \alpha_K - \lambda K, \qquad \frac{dR}{d\mathcal{A}} = \alpha_R - \lambda R, \qquad \frac{dY}{d\mathcal{A}} = \alpha_Y - \lambda Y,$$
$$\frac{dB}{d\mathcal{A}} = \alpha_B - \lambda B, \qquad \frac{dP}{d\mathcal{A}} = \alpha_P - \lambda P,$$

where $\mathcal{A}$ denotes the cumulative automation level.

10.3 Custodial Equilibrium

Setting derivatives to zero yields the equilibrium shares

$$K^* = \frac{\alpha_K}{\lambda}, \quad R^* = \frac{\alpha_R}{\lambda}, \quad Y^* = \frac{\alpha_Y}{\lambda}, \quad B^* = \frac{\alpha_B}{\lambda}, \quad P^* = \frac{\alpha_P}{\lambda},$$

with the normalization condition $\sum_i \alpha_i = \lambda$. The distribution of labor across janitorial domains is therefore determined by the relative entropy production rates of the corresponding civilizational substrates.

Theorem 10.2 (Fivefold Janitorial Equilibrium). *In a sufficiently automated economy, the occupational distribution converges to the custodial vector*

$$\mathbf{J}^* = (K^*, R^*, Y^*, B^*, P^*),$$

which represents the unique stable equilibrium of the redistribution dynamics.

Corollary 10.3 (Symmetric Case). *If the entropy production rates of the five domains are equal, so that $\alpha_i = \lambda/5$ for all i, the equilibrium is*

$$\mathbf{J}^* = \left(\tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}, \tfrac{1}{5}\right).$$

10.4 The Pentagonal Production Function

Combining all components of the analysis, total economic output in the custodial economy may be expressed as

$$Y = f(K, R, Y_\mu, B, P),$$

where Y_μ denotes the yogurt labor input (distinct from the output variable Y). The dependence of aggregate output entirely on the five maintenance factors constitutes a custodial analogue of the aggregate production function, in which the traditional inputs of capital and raw labor have been replaced by the five janitorial substrates of civilization. The full development of this production function is given in Section 12.

11 Custodial Boundaries and the Maintenance of Complex Systems

The custodial convergence described throughout this paper has a striking analogue in biological and informational systems. Complex structures capable of sustaining themselves over long periods almost always depend on boundary-maintaining processes that regulate the flow of energy, matter, and information.

In statistical physics and theoretical biology this role is often formalized through the concept of a *Markov blanket*, a set of nodes that separates the internal states of a system from its external environment while mediating all exchanges between them. A cell membrane performs an analogous function: it does not generate the metabolic activity of the cell, but regulates and maintains the conditions under which that activity can occur. The membrane removes waste, stabilizes chemical gradients, and protects internal processes from environmental disruption. In this sense the membrane performs a custodial function at the boundary of biological production.

A comparable structure appears in ecological systems. Forest canopies, soil layers, and microbial biofilms regulate exchanges between environmental compartments while stabilizing the conditions required for biological productivity. These examples suggest a general principle: whenever complex production processes arise, a complementary maintenance layer emerges whose primary role is to stabilize the system's boundary conditions.

The janitorial economy proposed in this paper is the macroeconomic analogue of these biological structures. Human labor increasingly operates at the boundaries of automated systems, performing functions analogous to those of cell membranes or ecological interfaces. Rather than generating production directly, custodial workers regulate the interfaces through which production systems interact with their environments. Teachers stabilize the informational boundary of educational systems. Programmers stabilize the software boundary of computational systems. Builders stabilize the structural boundary of physical infrastructure. Service workers stabilize the interaction boundary between automated systems and human users.

Figure 4 illustrates this boundary structure schematically.

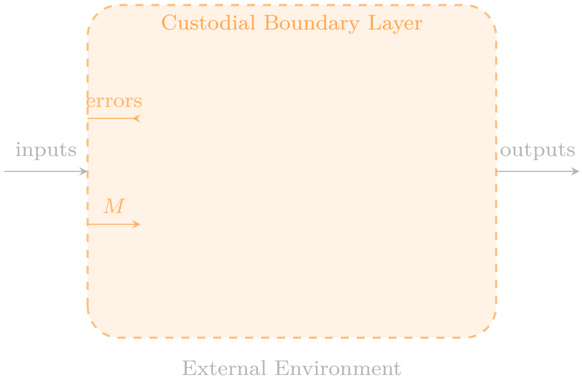

Figure 4: Schematic of the custodial boundary layer. The automated production system (inner rectangle) generates outputs and errors. The custodial boundary layer (dashed outer rectangle) intercepts errors, supplies corrective maintenance M, and regulates flows between the system and its environment. Human labor in the automated economy occupies this boundary layer.

This biological analogy clarifies the deeper interpretation of the Maintenance Convergence framework. The janitorial role is not merely a social artifact of automation but a structural feature of complex systems: as production systems become more sophisticated, the maintenance of their boundaries becomes increasingly important. The janitor therefore emerges as the economic counterpart of the cell membrane.

12 The Pentagonal Production Function

Classical growth theory models economic output as a function of capital and labor. The canonical representation is the Cobb–Douglas production function

$$Y = AK^{\alpha}L^{\beta},$$

in which aggregate output depends on physical capital K, labor L, and a technological productivity parameter A. In the custodial economy, however, productive activity increasingly occurs within automated systems. Human labor does not primarily generate output directly; rather, it maintains the infrastructures that enable production to occur. The relevant macroeconomic inputs are therefore not conventional labor categories but the five custodial regimes of Section 10.

Definition 12.1 (Pentagonal Production Function)**.** Define the custodial aggregate production function

$$Y = A\,K^{\alpha}R^{\beta}\hat{Y}^{\gamma}B^{\delta}P^{\epsilon},$$

where K, R, $\hat{Y}$, B, P denote employment shares in the kelp, rainforest, yogurt, bread, and paper domains respectively; $A > 0$ is the productivity of the automated technological substrate; and $\alpha, \beta, \gamma, \delta, \epsilon > 0$ are the output elasticities of the five custodial inputs. (The hat notation $\hat{Y}$ distinguishes the yogurt labor input from the output variable Y.)

Unlike traditional production functions, the pentagonal model assumes that output is generated primarily by automated processes embedded in complex infrastructures. Human labor contributes indirectly by stabilizing these infrastructures and preventing systemic failure. Each custodial domain therefore corresponds to the maintenance of a critical substrate of civilization: aquatic biogeochemical systems (kelp), terrestrial ecological networks (rainforest), microbial production systems (yogurt), agricultural and grain-processing systems (bread), and the informational substrate of coordination and knowledge (paper).

12.1 Balanced Custodial Equilibrium

In the symmetric case $\alpha = \beta = \gamma = \delta = \epsilon = \eta$, constant returns to scale requires $5\eta = 1$, so $\eta = 1/5$. With the symmetric equilibrium $K = R = \hat{Y} = B = P = 1/5$, aggregate output satisfies

$$Y^{*} = A\left(\frac{1}{5}\right)^{1} = \frac{A}{5}.$$

This configuration corresponds to a balanced custodial civilization in which human labor is evenly distributed across the five maintenance regimes.

12.2 Structural Implication

The Pentagonal Production Function captures the central thesis of this paper. In the automated economy, output depends less on the direct generation of goods by human labor and more on the stability of the infrastructures that produce those goods. Production becomes automated; maintenance becomes human. Figure 6 in Appendix C represents this equilibrium geometrically.

13 Potential Objections

The argument presented in this paper is intentionally stylized and invites several potential objections.

13.1 Automation Eliminates Labor Entirely

One might argue that sufficiently advanced automation eliminates the need for human labor altogether, in which case the janitorial economy would never emerge because machines would maintain themselves. The present framework implicitly assumes that complex systems generate entropy faster than automated maintenance mechanisms can remove it, so that human custodial intervention remains necessary. Formally, if system entropy evolves as

$$\frac{dE}{dt} = \lambda P - \mu M_a - \nu M_h,$$

where M_a denotes automated maintenance and M_h denotes human maintenance, the janitorial regime persists whenever $\lambda P > \mu M_a$. The Entropy Maintenance Principle (Theorem B.1) establishes this as the generic case under growing automated throughput.

13.2 Creative Professions Resist Convergence

A second objection holds that certain occupations—particularly creative or intellectual professions—may resist janitorial convergence. However, even creative domains increasingly rely on automated generative systems. As these systems expand, human participants shift toward roles involving selection, correction, and infrastructure management: the creative worker becomes a curator of automated outputs rather than their sole generator. The convergence theorem applies to this residual curatorial function.

13.3 Maintenance Is Not a Single Category

A third objection concerns the apparent heterogeneity of maintenance labor. Cleaning floors, debugging software, and preserving archives appear to be fundamentally different activities. The Fivefold Janitorial Model addresses this concern by partitioning maintenance labor into five ecological-industrial regimes corresponding to distinct substrates of civilization, while preserving the shared custodial structure that unifies them at the level of the formal model.

13.4 The Model Exaggerates a Structural Tendency

A final objection is that the model exaggerates for rhetorical effect. This observation is correct but incomplete. The exaggeration allows the model to isolate a structural tendency that might otherwise remain implicit in conventional labor economics discourse. The central claim—that technological systems redistribute human labor toward maintenance and repair—remains empirically recognizable even when the limiting case $(G, S, M) \to (0, 0, 1)$ is treated as a theoretical ideal rather than a literal prediction.

14 Discussion

The model developed in this paper formalizes a structural tendency observable across the history of technological societies. As productive systems become more autonomous, the remaining human role increasingly resembles infrastructure maintenance. This tendency is not incidental but follows necessarily from the dynamics of the automation operator A_t: generative labor decays exponentially, supervisory labor collapses through system stabilization, and maintenance labor absorbs all residual task mass.

Classical economic theory framed labor as the engine of value creation. The Maintenance Convergence framework inverts this picture. In the automated economy the human worker becomes the stabilizing agent that prevents complex technological systems from collapsing into disorder. The janitor provides the archetype of this role: janitorial labor is not generative in the classical sense, but preserves the functionality of systems that produce value elsewhere. In this sense the janitor stands in the same relation to the automated economy as the cell membrane stands to metabolic production: not a source of output, but the structural condition for its continuity.

The five custodial domains are not arbitrary classificatory choices. Each corresponds to a distinct class of entropy removal—oceanic biogeochemistry, terrestrial ecology, microbial fermentation, carbohydrate metabolism, and symbolic coordination—and together they span the full substrate of civilizational maintenance. The symmetry of the Fivefold Equilibrium (Corollary 10.2) suggests that in the long run these domains make approximately equal demands on human custodial effort, a claim that could in principle be tested against longitudinal employment data.

The model also raises a normative question that formal analysis cannot answer. If the custodial fixed point is structurally inevitable, the relevant question shifts from whether janitorial convergence occurs to what kind of custodial civilization is worth inhabiting. The section that follows takes up this question through a cultural case study.

15 Cultural Illustration: The Janitorial Equilibrium in Everyday Life

The custodial equilibrium derived in this paper also appears in contemporary cultural representation. A particularly striking example is provided by the film *Perfect Days* (2023), directed by Wim Wenders. The film follows Hirayama, a public toilet cleaner in Tokyo whose life is organized around an extremely stable daily routine. His work consists entirely of maintenance: cleaning shared public spaces, repairing small disturbances in the social environment, and ensuring that common infrastructure continues to function smoothly.

From the perspective of the Maintenance Convergence framework, Hirayama occupies the limiting occupational state

$$(G, S, M) \approx (0, 0, 1).$$

He does not produce goods in the conventional economic sense. He sustains the usability of an existing system. This places him formally at the custodial fixed point described in Appendix G: the state toward which all occupational trajectories converge under sustained automation.

What makes the film theoretically interesting is that it portrays the maintenance regime not as occupational degradation but as a stable and coherent form of life. Hirayama's routine—cleaning, photographing trees, listening to music on cassette, tending houseplants—exhibits the formal structure of custodial labor at every scale. Each activity involves the preservation or renewal of a small system in a state of order. The cassette player is maintained; the photographs preserve ephemeral states of the world; the plants are tended; the bathrooms are cleaned. The film presents these acts without irony.

The Fivefold Janitorial Model assigns Hirayama's labor primarily to the paper domain (symbolic and urban infrastructure), though his attentiveness to plants and urban ecology gives his custodial practice an ecological character that also touches the rainforest domain. More precisely, his role exemplifies the boundary-layer structure described in Section 11: he operates at the interface between the automated city and its human users, intercepting entropy and restoring functional order.

15.1 The Experiential Equilibrium

The mathematical framework treats the janitorial state as the limiting outcome of occupational dynamics. Within the model, convergence to $(0, 0, 1)$ appears as a structural inevitability: generative labor migrates into automated systems, supervisory labor collapses, and human effort concentrates on maintenance. The Entropy Maintenance Principle (Theorem B.1) further establishes that this maintenance demand grows proportionally with automated throughput, suggesting that the custodial layer expands continuously as civilization becomes more complex.

Perfect Days introduces a different register of analysis. Rather than presenting the janitorial role as a degraded residuum of generative work, the film portrays it as a *stable experiential equilibrium*: a configuration of daily life that is internally coherent, reproducible, and meaningful. Hirayama does not aspire to escape the custodial role; he inhabits it with care. His satisfaction does not depend on the generative significance of his output but on the quality of his attention to the task at hand.

This distinction has a precise formal correlate. In the dynamical model, the maintenance state $(0, 0, 1)$ is an attracting fixed point—a configuration to which the system returns after perturbation. The thermodynamic model similarly identifies maintenance as the steady state at which entropy production is balanced. The film suggests that this equilibrium has a subjective analogue: a mode of engagement with custodial work that is itself stable and self-reinforcing.

Definition 15.1 (Experiential Custodial Equilibrium)**.** An individual is said to occupy an experiential custodial equilibrium if their subjective orientation toward their work is stable under perturbation: that is, if small disruptions to routine are absorbed and repaired through the same custodial practices that constitute the work itself.

Hirayama's response to every disruption—a nephew who arrives unexpectedly, a broken bathroom fixture, a dying tree—is to perform a small act of maintenance and then return to routine. His equilibrium is robust precisely because maintenance is both his occupation and his adaptive strategy.

15.2 The Normative Inversion

The Maintenance Convergence Theorem establishes that all occupational trajectories converge to the janitorial fixed point. In the standard framing of labor economics this convergence appears as a loss: the displacement of skilled, generative, or creative work by routine maintenance. The progressive narrative of technological progress imagines the human worker as a producer whose role is diminished by automation.

Perfect Days performs a quiet inversion of this framing. By presenting the custodial fixed point as a possible site of human flourishing rather than economic failure, the film raises the possibility that the convergence result is not a tragedy to be resisted but a transition to be navigated with care.

This normative inversion does not depend on denying the mathematical structure of the model. It depends instead on recognizing that the formal result—$(G, S, M) \to (0, 0, 1)$—is underdetermined with respect to the question of what kind of life the maintenance state affords. The model specifies the task distribution of the equilibrium; it does not specify the phenomenology of inhabiting it.

The film therefore functions as a complement to the mathematical argument rather than a counterexample to it. Where the theorem establishes the structure of the limit, the film explores its experiential interior.

Remark 15.2. The standard welfare economics question "Is this equilibrium Pareto efficient?" may be less illuminating here than the phenomenological question: "Is this equilibrium humanly habitable?" *Perfect Days* argues, quietly and without sentimentality, that it is.

15.3 Domestic Custodianship and the Portability of Expertise

A further line of evidence supporting the Maintenance Convergence framework arises from the structure of everyday domestic life. In contemporary societies, nearly every household already performs a localized form of custodial governance. Individuals maintain their living environments by cleaning, organizing, repairing, and managing the small infrastructures that sustain daily life: appliances, food systems, documents, communication networks, and personal digital devices. These activities correspond almost exactly to the maintenance category M introduced in the task decomposition model. In effect, the home already functions as a miniature custodial economy—a small-scale instantiation of the janitorial equilibrium $(0, 0, 1)$ operating continuously outside any formal employment relationship.

Historically, professional labor took place primarily in centralized institutions—factories, offices, and schools—where expertise was institutionally embedded and geographically fixed. Let E_i denote the expertise vector associated with individual worker i. In the traditional institutional model, expertise was coupled to a fixed workplace W such that

$$E_i \subseteq W.$$

The worker's productive capacity existed only in relation to the particular tools, systems, and infrastructure housed within that institution.

Technological changes over the past several decades have begun to dissolve this coupling. The expansion of remote work, distributed computing, cloud credential systems, and portable digital identity allows many forms of expertise to detach from specific workplaces. In the distributed model, expertise becomes portable:

$$E_i \longrightarrow \mathcal{C}_i,$$

where $\mathcal{C}_i$ represents a cloud-mediated identity environment through which the worker accesses tools, collaborators, and information systems from any location. The institutional container W dissolves; what persists is the maintenance relationship between the worker and the automated systems they monitor.

Definition 15.3 (Custodial Node)**.** A *custodial node* is a spatially distributed locus at which a worker performs maintenance operations on remote automated systems. Formally, a custodial node is a pair $(i, \mathcal{C}_i)$ consisting of an individual worker and their associated cloud-mediated interface environment.

As expertise becomes portable and generative labor migrates into automated systems, the remaining human task is to maintain the interfaces between those systems and the environments in which they operate. In practical terms this means that a growing fraction of labor occurs within spaces that closely resemble domestic custodial environments. Workers monitor automated processes through remote dashboards, curate digital archives, stabilize communication networks, and correct machine-generated outputs—activities that are formally identical to the maintenance task M of the model, performed now from a home office rather than an institutional floor.

The boundary between professional infrastructure and household infrastructure therefore becomes increasingly blurred. A software engineer who debugs an automated pipeline from a home workstation is performing precisely the maintenance function that Theorem 4.1 predicts: no generative labor, no net supervision, only correction and upkeep. The physical environment is domestic; the task distribution is $(G, S, M) \approx (0, 0, 1)$.

Proposition 15.4. *In the limit of full portability, the set of custodial nodes* $\{(i, \mathcal{C}_i)\}$ *is coextensive with the residential geography of the labor force. Every home becomes a maintenance terminal of the automated economy.*

This proposition has a direct implication for the Fivefold Janitorial Model. If custodial nodes are distributed across residential space rather than concentrated in institutional facilities, then the five maintenance domains (kelp, rainforest, yogurt, bread, paper) no longer correspond to distinct spatial zones. Instead they correspond to distinct *functional interfaces* that may be managed from any location. Paper janitors maintain symbolic infrastructure remotely; yogurt janitors monitor fermentation bioreactors via sensor networks; rainforest janitors analyze ecological data from distributed field stations. The domestic sphere, in this reading, is not a private retreat from the economy but its distributed operational layer.

Perfect Days is again instructive here. Hirayama does not work from home, but his domestic routines and his professional routines are structurally identical: both consist in the careful maintenance of small systems. The cassette tapes he restores at home, the houseplants he waters, and the photographs he develops are domestic custodial acts whose structure is continuous with his professional cleaning. The film presents this continuity without comment, as though the distinction between domestic and professional maintenance were simply not operative for a worker who has fully inhabited the custodial equilibrium. From the formal perspective of the model, this makes complete sense: the task vector $(0, 0, 1)$ does not distinguish between institutional and residential contexts. Maintenance is maintenance.

The janitor may be the equilibrium state of the human economy.

16 Homeostatic Foundations of Custodial Behavior

The custodial framework proposed in this paper has a deeper grounding in the biology of living systems. All organisms must continuously maintain internal stability against environmental fluctuation. Survival depends not on the perpetual generation of new structures but on the maintenance of conditions that permit stable functioning. This is the principle of homeostasis, and it constitutes the oldest and most general instantiation of custodial behavior.

Three fundamental physiological requirements illustrate the principle at the level of individual organisms: sleep, nourishment, and thermal regulation. Each is not a productive act in the economic sense but a maintenance act: the restoration of a depleted or perturbed internal state to within viable bounds.

Definition 16.1 (Homeostatic State Vector)**.** Let the basic survival state of an organism be represented by

$$H = (S, \mathcal{E}, W),$$

where S denotes sleep stability, $\mathcal{E}$ denotes energetic reserves, and W denotes thermal regulation. Viable survival requires

$$H \in \Omega \ := \ [S_{\min}, S_{\max}] \times [\mathcal{E}_{\min}, \mathcal{E}_{\max}] \times [W_{\min}, W_{\max}].$$

The organism acts continuously to keep H within Ω: sleeping restores neurological and metabolic stability, eating replenishes energy reserves, and seeking or constructing shelter preserves the biochemical conditions necessary for cellular activity. In ecological terms, organisms organize their behavior around securing *affordances*—sleeping spaces, food sources, thermal shelter—that support these maintenance processes.

16.1 Collective Maintenance Infrastructure

Human societies extend individual homeostatic maintenance through collective organization. Housing systems stabilize sleeping environments; food production systems stabilize nutritional supply; architectural structures and energy networks stabilize thermal conditions. From this perspective, many social institutions are large-scale maintenance infrastructures whose function is to reproduce the basic affordances required for survival.

The formal parallel with the task decomposition model is immediate. At the level of the social organism, the five custodial domains of the Fivefold Janitorial Model correspond to distinct classes of collective homeostatic maintenance. Kelp and rainforest janitors maintain the ecological substrates that regulate planetary biogeochemistry. Yogurt and bread janitors maintain microbial and carbohydrate supply chains that stabilize nutritional reserves. Paper janitors maintain the informational infrastructure that coordinates all other maintenance activity.

Proposition 16.2. *The Fivefold Janitorial Model is a macroeconomic encoding of the five principal modes of collective homeostatic maintenance required for the survival of complex technological civilizations.*

16.2 Individual Custodianship as Homeostatic Behavior

At the individual scale, the custodial function appears not only in formal employment but in the everyday practices through which people sustain their own homeostasis. Preparing food, cleaning living spaces, repairing objects, managing warmth, and maintaining daily routines are all activities that keep the personal H-vector within Ω. These activities correspond precisely to the maintenance category M of the task decomposition model.

This observation provides a biological explanation for the robustness of the maintenance role across occupational domains. Custodial behavior is not a historically contingent feature of industrial economies; it is a structural feature of living systems. Any entity sufficiently complex to require homeostatic regulation will generate custodial labor as a necessary corollary of its own functioning.

Remark 16.3. The analogy may be extended to the level of cities and ecosystems. Urban ecology, infrastructure maintenance, and environmental stewardship are all forms of homeostatic regulation at the civilizational scale. The Maintenance Convergence Hypothesis can therefore be read as the prediction that technological automation drives the economy toward explicit expression of what biology has always required implicitly.

16.3 Implication for the Maintenance Convergence Hypothesis

The biological perspective resolves an apparent puzzle in the model. One might ask why maintenance labor should be structurally persistent rather than itself being automated. The homeostatic analysis answers: because the demand for maintenance is continuously regenerated by the entropy production of living and productive systems. Theorem B.1 establishes the proportionality $M^* = (\lambda/\mu)P$ in automated systems; the homeostatic argument extends this result to the biological level, showing that even in the absence of technological production, organisms must perform maintenance work to survive.

When technological systems assume responsibility for generative production, human activity shifts toward the maintenance of the environments that support those systems—which is precisely what the organism does anyway to sustain itself. The custodial economy predicted by the Maintenance Convergence framework therefore represents not an alien imposition of automation but an extension and amplification of the most basic property of living systems.

Living systems persist by maintaining the conditions that allow them to persist.

17 Conclusion

This paper has assembled a convergent body of evidence—dynamical, thermodynamic, renormalization-theoretic, categorical, empirical, biological, homeostatic, cultural, and domestic—for a single structural claim: sustained technological automation drives occupational structure toward a universal maintenance limit.

The Maintenance Convergence Theorem (Theorem 4.1) establishes this result at the level of the individual occupation. The Fivefold Janitorial Equilibrium (Section 10) characterizes the macroeconomic distribution of custodial labor. The Custodial Fixed Point theorem (Appendix G) identifies the limiting state as simultaneously dynamically stable, thermodynamically necessary, categorically terminal, and renormalization-group invariant. The cultural analysis of Section 15 adds a further dimension: the custodial fixed point is not only structurally inevitable but potentially experientially coherent. The analysis of domestic custodianship (Section 15.3) demonstrates that the janitorial equilibrium already operates at the household scale and that the portability of digital expertise extends it continuously into residential space. The homeostatic analysis (Section 16) grounds all of these arguments in biology: maintenance behavior is not a contingent feature of industrial economies but a structural necessity for any sufficiently complex living system. The Markov-blanket appendix (Appendix H) formalizes the boundary-layer interpretation, establishing that the custodial workforce occupies the probabilistic boundary separating automated production from its external environment.

Theorem 17.1 (Janitorial Convergence Law)**.** *Let $\mathcal{E}$ denote an economy undergoing sustained technological automation with aggregate level $\mathcal{A}$. As $\mathcal{A} \to \infty$, the occupational distribution of $\mathcal{E}$ converges to the universal maintenance state:*

$$\lim_{\mathcal{A}\to\infty} O = J = (0, 0, 1).$$

The traditional narrative of technological progress imagines a future in which machines perform labor while humans pursue creative or intellectual ends. The present analysis suggests a more austere outcome. As machines absorb responsibility for production, the human role converges to the maintenance of increasingly complex technological systems. The progressive vision in which automation liberates human creativity is not refuted by this analysis; it is simply not what the model predicts in the limit.

The cultural argument, however, introduces a qualification that the formal model cannot generate on its own. If the custodial fixed point is also an experiential equilibrium—a stable and humanly coherent mode of inhabiting the world—then the convergence result is not straightforwardly a loss. It is a structural transformation whose normative valence depends on questions the model leaves open: what practices of custodial care are possible within the maintenance regime, and under what social conditions do they become meaningful rather than degrading.

These questions belong to political economy and philosophy of labor rather than to mathematical economics. The present paper has prepared the formal ground on which they might be posed with greater precision.

In the fully automated economy, the janitor will not occupy the lowest rung of the economic hierarchy. The janitor will occupy the only rung. Whether that rung is a site of diminishment or of a quietly different kind of dignity is a question that no theorem can settle.

Every automated system eventually requires a human with a mop.

The question is whether the human with the mop is free.

A Renormalization of Occupational Structure

The maintenance convergence described in the main text may be understood through a coarse-graining argument analogous to the renormalization group procedures of statistical physics.

Consider a labor system composed of microscopic tasks $\tau_i \in \{\text{generate}, \text{supervise}, \text{maintain}\}$. Let an occupation be represented by a block $B = \{\tau_1, \tau_2, \ldots, \tau_n\}$. Define a coarse-graining operator $\mathcal{R}(B)$ that replaces the block with a single effective task according to the dominant functional activity. If automation removes a fraction α of generative tasks at each iteration, repeated coarse-graining defines a renormalization group trajectory $\mathcal{R}^k(T_O)$ in the space of task distributions.

Proposition A.1. *The maintenance state* $(0, 0, 1)$ *is a stable fixed point of the occupational renormalization group.*

Proof. Under coarse-graining, generative tasks shrink geometrically at each scale due to automation, while supervisory tasks decay through system stabilization. Maintenance tasks accumulate corrections at each scale. The transformation $(G, S, M) \mapsto (G', S', M')$ therefore flows monotonically toward $(0, 0, 1)$, which is invariant under $\mathcal{R}$. $\square$

The janitor appears as the infrared fixed point of labor dynamics: the effective description that survives repeated coarse-graining.

B Thermodynamic Interpretation

Technological systems generate entropy in the form of disorder, malfunction, and accumulation of errors and physical debris. Let E denote the operational entropy of a workplace and P the productive throughput of its automated systems. Entropy production satisfies

$$\frac{dE}{dt} = \lambda P - \mu M,$$

where $\lambda > 0$ governs the rate at which production generates disorder and $\mu > 0$ governs the rate at which maintenance labor removes it.

Theorem B.1 (Entropy Maintenance Principle)**.** *In steady state the required maintenance effort grows proportionally with productive throughput:*

$$M^* = \frac{\lambda}{\mu} P.$$

As automation increases P, maintenance labor M^* must increase in proportion.

This result provides a thermodynamic foundation for the main convergence argument. The janitor performs the physical removal of entropy produced by economic activity; the second law of thermodynamics guarantees that this role is never eliminated.

C Diagrammatic Representation of Professional Collapse

The convergence of occupational structure may be visualized as a contraction of the professional graph under the action of the janitorial functor F.

C.1 Occupational Convergence Map

The mapping $\Phi : \mathcal{O} \to \mathcal{J}$ acts as follows:

$$
\begin{aligned}
\text{Teacher} &\longrightarrow \text{Maintain learning systems} \\
\text{Programmer} &\longrightarrow \text{Maintain software systems} \\
\text{Designer} &\longrightarrow \text{Maintain generative design tools} \\
\text{Builder} &\longrightarrow \text{Maintain construction infrastructure} \\
\text{Customer service} &\longrightarrow \text{Maintain support systems}
\end{aligned}
$$

Taking the limit yields

$$\forall\, O \in \mathcal{O}, \quad \Phi(O) = J = (0, 0, 1).$$

C.2 Schematic Visualization

Figure 5 presents a schematic of the convergence process. The upper panel represents a diversified professional economy. The lower panel illustrates the predicted long-run equilibrium.

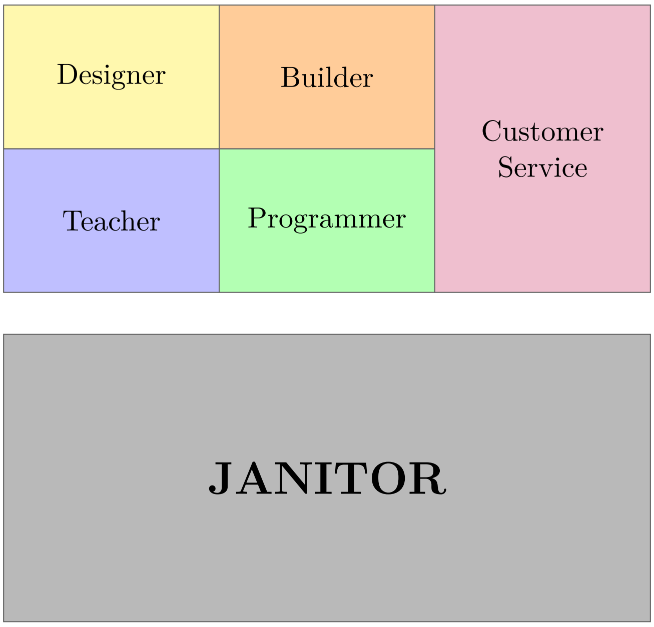

Figure 5: Schematic visualization of occupational convergence under automation. *Upper*: a diversified professional economy with distinct task compositions. *Lower*: the predicted long-run equilibrium in which all occupations collapse into the universal maintenance role $(0, 0, 1)$.

C.3 The Fivefold Custodial Pentagon

Figure 6 represents the structure of the maintenance economy at the custodial equilibrium $\mathbf{J}^*$. Each vertex corresponds to one of the five fundamental maintenance domains.

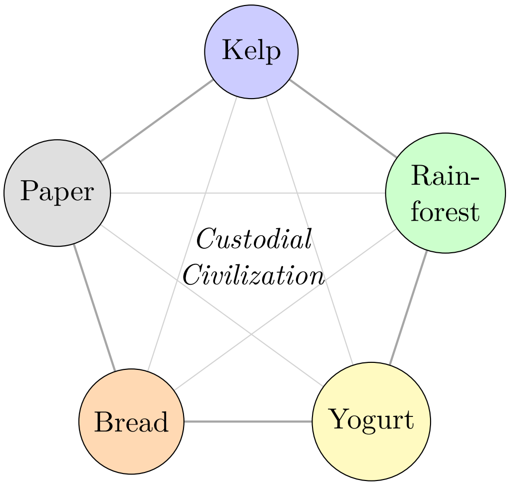

Figure 6: The Fivefold Janitorial Model at custodial equilibrium $\mathbf{J}^*$. Each vertex represents a fundamental maintenance domain of the automated economy: aquatic systems (kelp), ecological infrastructure (rainforest), microbial ecosystems (yogurt), industrial food systems (bread), and symbolic knowledge systems (paper). Internal edges indicate inter-domain dependencies.

D Stability Analysis of the Maintenance Equilibrium

Consider the task vector $T = (G, S, M)$ with dynamics given by equations (1)–(3). The equilibrium of interest is $T^* = (0, 0, 1)$. To analyze stability, we linearize the system about T^* by writing $G = 0 + g$, $S = 0 + s$, $M = 1 - g - s$ for small perturbations g, s. The linearized equations are

$$\frac{d}{dt}\begin{pmatrix} g \\ s \end{pmatrix} = \begin{pmatrix} -\alpha & 0 \\ \beta & -\gamma \end{pmatrix}\begin{pmatrix} g \\ s \end{pmatrix}.$$

The eigenvalues of the coefficient matrix are $-\alpha$ and $-\gamma$. Since both are strictly negative, the equilibrium is asymptotically stable in the (G, S) plane; the M component follows by the normalization constraint.

Proposition D.1. *The maintenance equilibrium $T^* = (0, 0, 1)$ is a stable node of* *the automation dynamics for all $\alpha, \gamma > 0$. Trajectories approach T^* exponentially at rates $e^{-\alpha t}$ and $e^{-\gamma t}$.*

Figure 7 depicts the phase portrait of the projected (G, S) system, confirming the stable-node character of the origin.

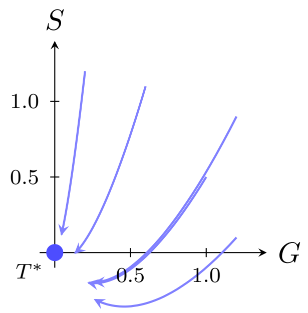

Figure 7: Phase portrait of the projected dynamics in the (G, S) plane. All trajectories converge to the stable node at the origin $T^* = (0, 0)$ (corresponding to the full maintenance state $(0, 0, 1)$), confirming asymptotic stability for all $\alpha, \gamma > 0$.

E Entropy Balance in Automated Systems

Let $E(t)$ denote the operational entropy of a production system. Productive throughput P generates entropy at rate λP; maintenance labor M removes entropy at rate μM. The entropy balance is

$$\frac{dE}{dt} = \lambda P - \mu M.$$

Steady state requires $\lambda P = \mu M$, giving the maintenance demand

$$M^* = \frac{\lambda}{\mu} P.$$

As automation raises P substantially, M^* must rise proportionally. This result, stated as Theorem B.1 in Appendix B, has a geometric interpretation.

F The Janitor as Terminal Object

We develop the categorical argument in full. Let $\mathcal{E}$ denote the category of economic roles. Objects are occupations or socially differentiated forms of labor;

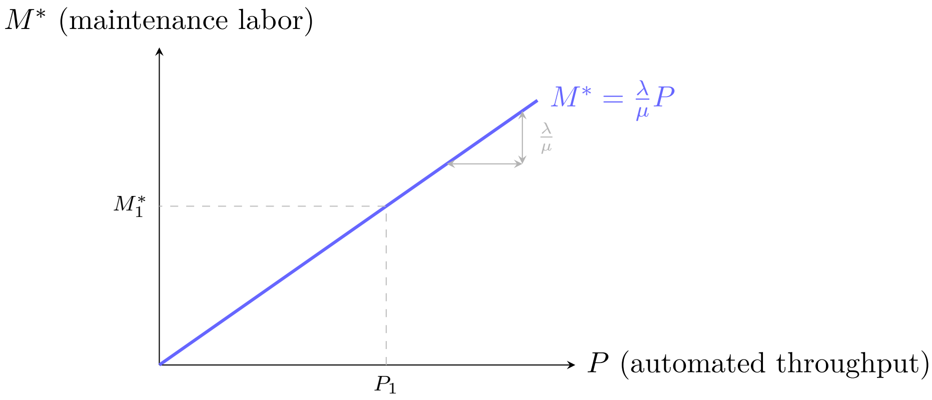

Figure 8: Linear scaling of maintenance labor M^* with automated productive throughput P. The slope λ/μ measures the ratio of entropy production rate to entropy removal efficiency. As automation raises P, the required custodial effort grows in proportion.

morphisms represent admissible structural transformations—automation of generative tasks, conversion of craft into supervisory work, and collapse of supervision into maintenance.

Define a distinguished object $J \in \mathrm{Ob}(\mathcal{E})$, the *janitor*, understood not as a narrow job title but as the abstract economic role whose dominant function is the maintenance of system conditions, the removal of local entropy, and the preservation of operational continuity.

Definition F.1 (Terminal Object)**.** An object T in a category $\mathcal{C}$ is *terminal* if for every object $X \in \mathrm{Ob}(\mathcal{C})$ there exists a unique morphism $X \to T$.

Proposition F.2. *For every occupation $O \in \mathrm{Ob}(\mathcal{E})$, there exists a morphism $\phi_O : O \to J$.*

Proof. By Theorem 4.1, the task composition of O satisfies $\lim_{t\to\infty} T_O(t) = (0, 0, 1)$. The generative component is automated, the supervisory component collapses, and the maintenance component absorbs the surviving functional content of the role. This asymptotic projection defines the required structural map ϕ_O. □

Proposition F.3. *For every occupation $O \in \mathrm{Ob}(\mathcal{E})$, the morphism $\phi_O : O \to J$ is unique up to custodial equivalence.*

Proof. Any two maps $\phi_O, \psi_O : O \to J$ represent long-run convergence under automation. Since J is defined as the role of pure maintenance, any two such maps differ only in contingent details—tool choice, institutional setting, substrate of entropy removal—that are identified under the equivalence relation that preserves maintenance function across different environments. Hence $\phi_O \sim \psi_O$. □

Theorem F.4. *The janitor J is a terminal object in the category $\mathcal{E}$ of economic roles.*

Proof. Immediate from the previous two propositions. $\square$

F.1 Functorial Formulation

The construction may be expressed functorially. Define the maintenance collapse functor

$$\mathcal{M} : \mathcal{E} \to \mathbf{1},$$

where $\mathbf{1}$ is the terminal category with one object and one morphism. Equivalently, define

$$F : \mathcal{E} \to \mathcal{J},$$

where $\mathcal{J}$ is the full subcategory generated by J. Then $F(O) = J$ for all $O \in \mathrm{Ob}(\mathcal{E})$, and every occupational history factors through J in the long run.

F.2 Universal Property and Occupational Convergence

The terminality of J means that the janitor satisfies a universal property: any occupation becomes intelligible, in the limit, as a special case of custodial work. Teaching becomes the maintenance of learning environments; programming the maintenance of software environments; design the maintenance of aesthetic-selection environments; construction the maintenance of structural environments; customer service the maintenance of interaction environments.

The janitor is not merely one profession among others. The janitor is the universal receiver of occupational convergence: the object toward which every arrow in $\mathcal{E}$ is eventually directed.

Since every object of $\mathcal{E}$ admits a unique morphism into J, the multiplicity of professions is a finite-time phenomenon. At sufficient temporal or technological scale, occupational diversity is revealed to be a family of morphisms converging upon the same terminal object.

Every job has its arrow to the janitor.

G The Custodial Fixed Point

Combining the dynamical (Section 4), thermodynamic (Appendix B), stability (Appendix D), and categorical (Appendix F) arguments yields the complete characterization of the limiting occupational state:

$$(G, S, M) \;\longrightarrow\; (0, 0, 1).$$

In macroeconomic terms this is the custodial fixed point of civilization: the state in which all human labor is devoted to maintaining the automated systems that generate output.

Theorem G.1 (Custodial Fixed Point)**.** *The state* $(0, 0, 1)$ *is simultaneously (a) the asymptotically stable equilibrium of the automation dynamics, (b) the infrared fixed point of the occupational renormalization group, (c) the thermodynamic steady state of maximum entropy production, and (d) the terminal object of the category of professions.*

All sufficiently advanced economies converge to the maintenance of their own complexity.

H Homeostasis and Custodial Markov Boundaries

Section 11 introduced the biological analogy between custodial labor and Markov-blanket boundary processes. This appendix develops the formal structure of that analogy.

Definition H.1 (Markov Blanket)**.** Let $X = (I, B, E)$ denote the state of a system partitioned into internal states I, boundary states B, and external states E. The set B is a *Markov blanket* for I if

$$P(I, E \mid B) = P(I \mid B)\, P(E \mid B),$$

that is, if I and E are conditionally independent given B.

In biological organisms, B corresponds to the cell membrane or other regulatory interfaces. The boundary regulates flows of matter and energy so that the internal states remain within the viable region Ω defined in Section 16.

H.1 Custodial Labor as Boundary Regulation

The occupational task vector $C = (G, S, M)$ is related to the Markov partition as follows. Automated systems perform internal productive activity I; the external environment corresponds to E; custodial workers regulate the interface B. Under the automation dynamics, the maintenance component M converges to 1, meaning that human labor concentrates entirely in the boundary layer.

Proposition H.2. *In the limit $(G, S, M) \to (0, 0, 1)$, the human labor force occupies the Markov blanket B of the automated economy.*

H.2 Stability Condition

For the system to remain viable, the custodial boundary must remove perturbations at least as fast as they are introduced. Let D denote the rate of disturbance generation and R the rate of maintenance repair. The viability condition is

$$R \ \geq \ D.$$

In highly automated economies, increasing production complexity raises D continuously. The Entropy Maintenance Principle (Theorem B.1) establishes that the required repair rate $R = M^*$ grows proportionally with automated throughput. The custodial boundary must therefore expand as civilization becomes more complex, a conclusion that reinforces the Maintenance Convergence Theorem from a different direction.

H.3 Summary

The Markov-blanket interpretation unifies the biological and economic arguments of the paper. In living organisms the membrane maintains the conditions for metabolic production. In technological civilizations the custodial workforce maintains the conditions for automated production. The convergence of human labor toward the maintenance role is therefore not merely an economic phenomenon; it is the macroeconomic expression of a structural property shared by all complex adaptive systems.

Civilizations survive by maintaining their boundaries.

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